Stochastic Modelling of Statistical Arbitrage with Ornstein–Uhlenbeck Processes: A Mean-Reversion Pairs Trading Strategy on NSE Bank Stocks

MAC F244 Stochastic Calculus and Applications to Finance — Course Project

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Abstract

We model the log-price spread of a pair of cointegrated stocks as an Ornstein–Uhlenbeck (OU) process, derive its solution and transition law with Itô's lemma, and estimate it through the exact AR(1) discretisation on rolling 120-day windows. Monte Carlo experiments confirm the theoretical moments and show that Euler–Maruyama becomes unstable for $\theta\Delta t \geq 2$ while the exact scheme does not. They also show that OLS overestimates the reversion speed θ by about 100% on 120-day windows. A z-score trading rule, backtested out of sample on six Nifty Bank stocks (Jan 2024–Sep 2026, 10 bps per leg), returns -0.6% net for an equal-weight portfolio of three pairs. Thresholds from the OU first-passage-time problem improve the one pair that passes Engle–Granger ($+10.0\%$ vs. -4.0%) but not the portfolio. The OU machinery is sound. The trading edge it produces here is small, fragile, and dominated by transaction costs and parameter instability.

1 Introduction and data

Pairs trading [4] bets that the price gap between two economically linked stocks will close. If the gap behaves like a mean-reverting diffusion [1], stochastic calculus gives its full law and tells us when to trade. We use daily adjusted closes of six Nifty Bank constituents (HDFC Bank, ICICI Bank, Axis Bank, Kotak Mahindra Bank, SBI, IndusInd Bank) from Yahoo Finance: 1,915 aligned trading days with no missing values. All model choices use the *training* period (Jan 2019–Dec 2023, 1,235 days). The *test* period (Jan 2024–Sep 2026, 680 days) is used only for evaluation.

2 Mathematical model

Spread. With $y_t = \ln P_t^A$ and $x_t = \ln P_t^B$, both $I(1)$, OLS on the training data gives $y_t = \alpha + \beta x_t + \varepsilon_t$. The spread $X_t = y_t - \beta x_t$ is stationary if and only if the pair is cointegrated. We test this with the Engle–Granger (EG) two-step test and MacKinnon critical values [2].

OU dynamics. On a filtered space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ with Brownian motion W_t,

$$dX_t = \theta(\mu - X_t)\,dt + \sigma\,dW_t, \qquad \theta, \sigma > 0. \tag{1}$$

Itô's lemma applied to $f(t, X_t) = e^{\theta t} X_t$ gives $d(e^{\theta t} X_t) = \theta\mu e^{\theta t} dt + \sigma e^{\theta t} dW_t$, since $f_{xx} = 0$. Integrating,

$$X_t = \mu + (X_0 - \mu)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)}\,dW_s. \tag{2}$$

The stochastic integral has a deterministic integrand, so it is Gaussian with mean 0. By the Itô isometry [6] its variance is $\sigma^2 \int_0^t e^{-2\theta(t-s)} ds$. Hence

$$X_t \mid X_0 \sim \mathcal{N}\Big(\mu + (X_0 - \mu)e^{-\theta t},\ \tfrac{\sigma^2}{2\theta}(1 - e^{-2\theta t})\Big) \xrightarrow{t\to\infty} \mathcal{N}(\mu, \sigma_{eq}^2), \quad \sigma_{eq} = \frac{\sigma}{\sqrt{2\theta}}. \tag{3}$$

The expected deviation halves after $t_{1/2} = \ln 2/\theta$. With $\Delta t = 1/252$, this is $252 \ln 2/\theta$ trading days.

Exact discretisation. Setting $t = \Delta t$ in (2) gives an *exact* Gaussian AR(1):

$$X_{t+\Delta t} = a + bX_t + \eta_t, \quad b = e^{-\theta \Delta t},\ a = \mu(1-b),\ \eta_t \sim \mathcal{N}(0, \sigma_{eq}^2(1-b^2)). \tag{4}$$

Inverting the OLS estimates $(\hat{a}, \hat{b}, s^2)$ gives $\hat{\theta} = -\ln \hat{b}/\Delta t$, $\hat{\mu} = \hat{a}/(1-\hat{b})$, $\hat{\sigma}_{eq}^2 = s^2/(1-\hat{b}^2)$ and $\hat{\sigma} = \hat{\sigma}_{eq}\sqrt{2\hat{\theta}}$. These are well defined only for $0 < \hat{b} < 1$. For comparison, Euler–Maruyama gives $X_{n+1} = X_n + \theta(\mu - X_n)\Delta t + \sigma\sqrt{\Delta t}\, Z_n$. Its amplification factor is $g = 1 - \theta\Delta t$, so it is stable only when $|g| < 1$, i.e. $\theta\Delta t < 2$. Even then its stationary variance is $\sigma^2/[\theta(2-\theta\Delta t)]$ rather than $\sigma^2/(2\theta)$.

Trading rule and first-passage times. The standardised spread $Z_t = (X_t - \mu)/\sigma_{eq}$ satisfies $dZ_t = -\theta Z_t\, dt + \sqrt{2\theta}\, dW_t$, with generator $\mathcal{A}f = \theta f'' - \theta z f'$. We go short the spread when Z reaches $+s_e$ (long at $-s_e$) and close when it reverts to $s_x < s_e$. The expected holding time $T(z)$ solves Dynkin's equation $\mathcal{A}T = -1$ with $T(s_x) = 0$, which gives [3]

$$\mathbb{E}[T_{\text{trade}}] = \frac{\sqrt{2\pi}}{\theta} \int_{s_x}^{s_e} e^{u^2/2}(1 - \Phi(u))\, du, \qquad \mathbb{E}[T_{\text{cycle}}] = \frac{\pi}{2\theta}\Big[\operatorname{erfi} \tfrac{s_e}{\sqrt{2}} - \operatorname{erfi} \tfrac{s_x}{\sqrt{2}}\Big]. \tag{5}$$

The cycle time adds the two-sided wait for $|Z|$ to return to s_e. Maximising the expected return rate $\big[(s_e - s_x)\sigma_{eq}/(1+|\beta|) - c\big]/\mathbb{E}[T_{\text{cycle}}]$ with round-trip cost c gives "FPT-optimal" thresholds (s_e^*, s_x^*).

3 Numerical methodology

1. **Pair selection.** We run EG tests, OLS hedge ratios and full-sample AR(1) fits for all $\binom{6}{2} = 15$ pairs on training data, keeping pairs with $5 \le t_{1/2} \le 60$ days.
2. **Rolling estimation.** β is fixed from training. $(\theta, \mu, \sigma, \sigma_{eq})$ are re-estimated with (4) on every 120-day window (1,796 windows per pair). Windows with $\hat{b} \notin (0,1)$ are flagged invalid, never clipped. The signal is $s_t = (X_t - \hat{\mu}_t)/\hat{\sigma}_{eq,t}$, which uses data only up to day t.
3. **Simulation.** With $\theta = 13.52$, $\sigma = 0.25$ and $\mu = 0.15$, we (i) drive the exact and Euler schemes with the *same* normals for $\Delta t \in \{1, 5, 10, 21, 63\}$ days (500 paths); (ii) compare 10,000-path Monte Carlo moments with (3); (iii) simulate from known (θ, σ) and re-estimate to measure estimator bias.
4. **Backtest.** A signal at the close of day t trades at that close and earns the day-$(t{+}1)$ return. Positions have unit gross exposure: weights $\pm 1/(1+|\beta|)$ and $\mp\beta/(1+|\beta|)$, spread return $(r_t^A - \beta r_t^B)/(1+|\beta|)$. The baseline uses $s_e = 2$, $s_x = 0.5$, a stop-loss at $|s_t| \ge 3.5$ and 10 bps per leg per side (40 bps round trip). The portfolio puts 1/3 of capital in each pair. Results are compared with FPT-optimal thresholds and buy-and-hold.

4 Results

Pair selection (Table 1). Correlation does not imply cointegration. The most correlated pair, ICICI–SBI ($r = 0.95$), is far from cointegrated (EG $p = 0.20$). Only HDFC–Kotak passes EG at 5% ($p = 0.048$, $\beta = 1.09$, $t_{1/2} = 31$ days). We keep the two next-best pairs, Kotak–SBI and HDFC–SBI, as *borderline* comparisons ($p = 0.06$ and 0.10). Their residuals pass a plain ADF test, but plain ADF critical values are too lenient for estimated residuals.

Table 1: In-sample (2019–2023) statistics. θ in yr^{-1}; half-life in trading days.

Pair	Corr. r	β	EG p	$\hat{\theta}$	$t_{1/2}$	Role
HDFC Bank – Kotak	0.908	1.093	**0.048**	5.64	31.0	primary (cointegrated at 5%)
Kotak – SBI	0.775	0.289	0.060	4.57	38.2	borderline comparison
HDFC Bank – SBI	0.879	0.394	0.097	4.01	43.6	borderline comparison
ICICI – SBI	0.948	0.874	0.202	3.10	56.4	rejected (highest correlation)

Numerical validation (Fig. 1). The Monte Carlo mean and variance follow (3) to within 0.7% and 1.9% at every checkpoint. The terminal variance is 0.002293 against the theoretical $\sigma_{eq}^2 = 0.002311$. Coupled Euler error grows with the step: RMSE is 1.3×10^{-3} at daily steps and 4.7×10^{-2} at monthly

steps ($g = -0.13$, oscillatory but stable). At quarterly steps $\theta\Delta t = 3.38 > 2$ and the error explodes (RMSE 4.04, $g = -2.38$), while the exact scheme is unaffected for any Δt. Panel (c) is the result that matters most in practice. On 120-day windows OLS roughly *doubles* θ ($\mathbb{E}\hat{\theta} = 20.1$ vs. 10). This is the downward small-sample bias of $\hat{b}$ (Hurwicz/Marriott–Pope, $\mathbb{E}\hat{b} - b \approx -(1+3b)/N$ [5]) passed through $-\ln \hat{b}$. The bias falls to +7% at $N = 1235$, while $\hat{\sigma}$ is unbiased throughout. The real data are consistent with this: the rolling estimates for HDFC–Kotak have mean $\hat{\theta} = 16.3$ (median 13.5, median half-life 12.9 days) against the full-sample 5.64. Because $\hat{\sigma}_{eq} = \hat{\sigma}/\sqrt{2\hat{\theta}}$, an inflated $\hat{\theta}$ shrinks $\hat{\sigma}_{eq}$ and pushes $|s_t|$ outward.

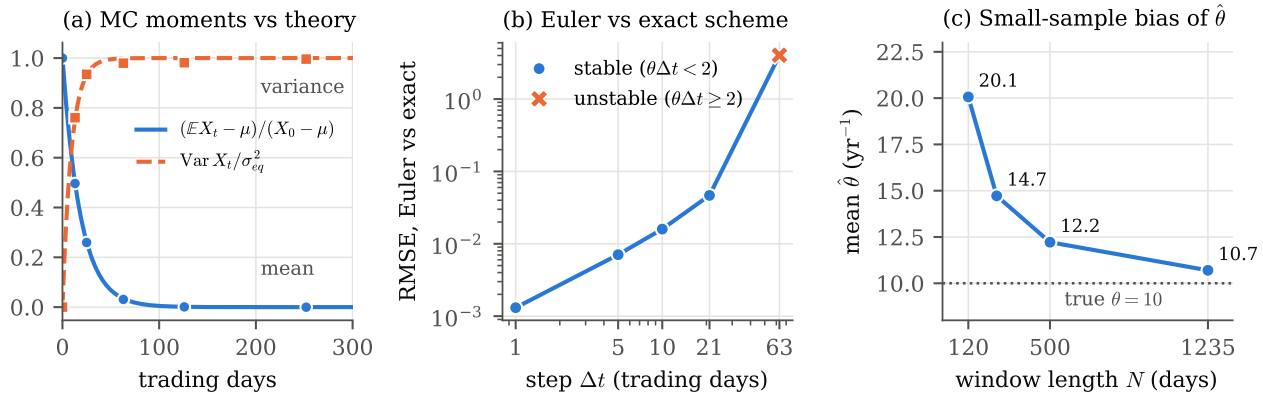


Figure 1: (a) Monte Carlo mean decay and variance build-up (markers) against the closed forms (3) (lines), both normalised to $[0, 1]$. (b) Pathwise RMSE between coupled Euler and exact paths; the quarterly step violates $\theta\Delta t < 2$. (c) Mean OLS estimate of θ against window length (true $\theta = 10$, $\sigma = 0.25$).

Out-of-sample trading (Table 2, Fig. 2). With the baseline rule the portfolio made 18 trades and finished at −0.58% net (+1.84% gross). It had low volatility (2.5% p.a.) and a 3.6% maximum drawdown, but did not beat cash. The cointegrated primary pair traded only three times and lost 4.0%. The only profitable pair, HDFC–SBI (+13.0%, 8 of 9 trades won), is a *borderline* pair, and its gains are not robust (see below). The FPT optimiser returns $s_e^* = 1.25$ (the lower bound of the search) and $s_x^* = 0$ for all three pairs. On the primary pair this raises the net return to +10.0% over 13 trades, with an average hold of 38 days against the 47 days predicted by (5) with $\hat{\theta} = 5.64$. On the borderline pairs it loses heavily, so the FPT portfolio is *worse* (−7.6%). For scale, an equal-weight bank basket returned +9.6% with 17% volatility.

Table 2: Out-of-sample performance, 1 Jan 2024–30 Sep 2026 (680 days), net of 10 bps per leg. Sharpe uses $r_f = 0$; win rate is on net trade P&L.

Rule (s_e/s_x)	Pair	Trades	Win	Hold (d)	Net return	Ann. vol	Sharpe	Max DD
Baseline (2.0 / 0.5)	HDFC–Kotak	3	0%	25.7	−3.99%	2.8%	−0.54	4.8%
	HDFC–SBI	9	89%	8.8	+13.02%	5.1%	0.92	7.0%
	Kotak–SBI	6	33%	13.5	−9.94%	5.2%	−0.73	10.0%
	Portfolio	18	56%	13.2	**−0.58%**	2.5%	−0.09	3.6%
FPT-optimal (1.25 / 0)	HDFC–Kotak	13	69%	38.2	+9.96%	9.7%	0.37	7.3%
	HDFC–SBI	11	45%	46.0	−20.31%	12.1%	−0.67	27.7%
	Kotak–SBI	14	50%	20.1	−12.64%	9.8%	−0.50	18.3%
	Portfolio	38	55%	33.8	**−7.63%**	6.3%	−0.46	10.8%
Buy & hold	6-bank basket	–	–	–	+9.63%	17.0%	0.20	18.0%

Robustness. (i) *Costs:* the portfolio's gross edge of 1.84% is used up at about 7.5 bps per leg (net +0.02%). At 10 bps the cost drag (2.4% of capital) is larger than the gross profit. (ii) *Parameter instability:* when a window is close to a unit root ($\hat{b} \to 1$), $\hat{\mu} = \hat{a}/(1-\hat{b})$ is ill-conditioned. On 1 Mar 2024 HDFC–SBI had $\hat{\mu} = -7.6$ while the spread was 3.93, giving $s_t = 15.7$. In 23% of that pair's test windows, $\hat{t}_{1/2} > 60$ days or the window is invalid. Applying the selection criterion $t_{1/2} \leq 60$ to the rolling windows cuts HDFC–SBI from 9 trades to 2 and moves the portfolio to +0.29%. (iii) *Stop-loss:*

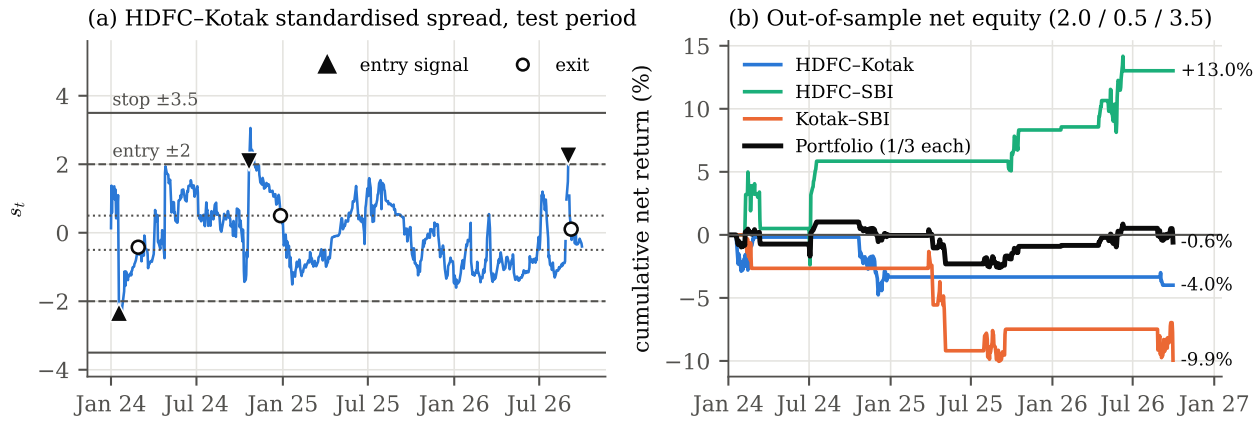


Figure 2: (a) Rolling standardised spread of the primary pair over the test period, with entry (±2), exit (±0.5) and stop (±3.5) bands. (b) Cumulative net return of each pair and the equal-weight portfolio under the baseline rule.

all five stop-loss exits on HDFC–SBI were triggered by jumps in $(\hat{\mu}_t, \hat{\sigma}_{eq,t})$ rather than by the spread diverging, and all five closed trades that were in profit. Removing the stop raises that pair to +25.1% and the portfolio to +2.1%. So the sign of the result depends on choices that cannot be fixed in advance from the training data.

5 Conclusions

- **The model works as stochastic calculus predicts.** The Itô solution, the Gaussian transition law and the exact AR(1) map are confirmed numerically. The exact scheme should be used for both simulation and estimation, since Euler adds bias and can become unstable.
- **Estimation is the weak link.** On short windows, OLS doubles θ, and near unit roots $\hat{\mu}$ and $\hat{\sigma}_{eq}$ are unreliable. The trading signal inherits both problems. Bias-corrected or maximum-likelihood estimators, longer windows, and a half-life filter on rolling estimates are natural next steps.
- **No reliable out-of-sample edge after costs.** A disciplined rule with no look-ahead lost 0.6% over 2.7 years while cash earned about 18%. FPT-optimal thresholds helped only on the one genuinely cointegrated pair. That pair was selected at $p = 0.048$ out of 15 tests, which is not significant after a multiple-testing correction. Cointegration estimated on 2019–2023 did not carry over reliably into 2024–2026.

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